Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

ML engineer with a **full predictive modeling and production deployment lifecycle** - built CrashClarity (85%+ accuracy Random Forest pipeline on 100K+ records), AgentProbe's LLM evaluation engine with automated experimentation across providers, and distributed data pipelines processing 1M+ records at Entrata. Now building at the intersection of LLMs and intelligent decisioning: multi-provider content generation, automated eval runs, and production observability infrastructure. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Demo: *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Predictive Modeling | ML Lifecycle (Data → Prod) | Automated Experimentation | LLM Content Generation | Feature Engineering | ML Infrastructure & Deployment | LangChain / LangGraph | RAG Pipelines | Python / FastAPI | Multi-Provider LLM Routing | Large-Scale Data Pipelines | AWS / Docker / CI/CD

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Built **ML and data processing infrastructure** on AWS (S3, Glue, EC2) handling **1M+ records** with automated quality checks - the production ML deployment discipline this role requires.
- Improved production query performance by **30%** by architecting PostgreSQL schemas across 20+ projects; maintained **99.5% uptime** for 6+ months via TDD/BDD quality gates.
- Built **20+ production modules** (Python, TypeScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users**; cut microservices execution time by **50%** via Docker containerization.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Delivered Python and database mentoring for 30+ undergraduates; translated complex technical concepts for non-technical audiences.

PROJECTS

AgentProbe: LLM Evaluation & Automated Experimentation Platform OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama

Production ML system built around automated experimentation and LLM quality decisioning. **Automated experimentation engine**: runs multi-provider eval experiments (Anthropic, OpenAI, Groq, Ollama) with 5 metrics per run - equivalent to systematic A/B testing across model and prompt variants. **LLM content generation**: multi-provider routing with quality scoring, prompt engineering, and output evaluation - directly applicable to content generation for 1:1 personalization. **Predictive failure detection**: anomaly flags surface regressions before they compound. **15-endpoint FastAPI production backend** with Docker, CI/CD, observability dashboard. *github.com/venkatalakshmiparimala/AgentProbe*

CrashClarity: Predictive Modeling Pipeline on 100K+ Records ML LIFECYCLE

Python · R · Random Forest · Logistic Regression · Feature Engineering · Statistical Analysis · GitHub Pages

Full ML lifecycle on UK road accident dataset: problem definition → feature engineering → model selection → training (Random Forest vs Logistic Regression) → evaluation → deployment. **85%+ accuracy**; interactive dashboard presenting insights to non-technical stakeholders. Demonstrates the end-to-end predictive modeling workflow at the core of AI Decisioning.

AI on FHIR: NLP-Driven Medical Records Query System

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

NLP pipeline mapping natural language queries to structured FHIR APIs via LLMs, BERT, embeddings. HIPAA-compliant RBAC. **Reduced retrieval time ~60%**. *github.com/venkatalakshmiparimala/AI_on_FHIR*

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, data pipelines at scale

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

ML & Modeling: PyTorch, TensorFlow, scikit-learn, Random Forest, Logistic Regression, Feature Engineering, NumPy, Pandas, BERT, spaCy, Hugging Face, Statistical Modeling, Experimental Design

LLM & AI: LangChain, LangGraph, Multi-Provider LLM (Anthropic, OpenAI, Groq, Ollama), RAG Pipelines, ChromaDB, Prompt Engineering, LLM Evaluation, Automated Experimentation, FastAPI

Data & Cloud: AWS (S3, EC2, RDS, Glue), Azure, PostgreSQL, MySQL, NoSQL, Docker, CI/CD (GitHub Actions), RabbitMQ

Testing & Quality: TDD/BDD, Integration Testing, Git, JIRA, Postman, Agile/Scrum